VIT

Vellore Institute of Technology

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School of Computer Science Engineering and Information Systems

Fall Semester 2026–2027

SOFTWARE ENGINEERING LABORATORY

LastPlate

Dynamic Food Delivery & Real-Time Surplus Food Rescue Platform

Assessment 9: User Interface Design and Human-Computer Interaction Report

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Course: Software Engineering (BCSE301P) | Assessment-9 [10 Marks]

Name	Registration No
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Solo Full-Stack Architect & User Interface Designer

GitHub repository	github.com/ahlawatansh/LastPlate
Live web application (Vercel)	lastplate.vercel.app
Module Evaluated	Multi-Role Responsive UI & Dynamic Surplus Interaction Design
UI/UX Design Frameworks	React 19, Tailwind CSS v4, Motion, Lucide Icons, Mobile-First Ergonomics

Under the Guidance of

Course Instructor & Faculty in Charge

Department of Software and Systems Engineering

School of Computer Science Engineering and Information Systems

1 Abstract and Executive Summary

LastPlate is an enterprise-grade food delivery and real-time surplus rescue application engineered to solve commercial food wastage through dynamic clearance auctions. This report presents the comprehensive **User Interface (UI) and Human-Computer Interaction (HCI) Design** for the LastPlate multi-role platform, adhering strictly to established usability heuristics, cognitive ergonomics, and modern responsive design standards.

The interface spans four role-specific portals (Customer, Restaurant Partner, Delivery Fleet, and Platform Administrator) unified under a coherent design system powered by **React 19, Tailwind CSS v4, Motion physics, and Lucide vector iconography**. The interface addresses complex real-time operational flows—including dynamic decaying discount counters, sliding basket management, GPS order tracking, multi-rail checkout, and kitchen prep forecasting—while maintaining strict simplicity, visual hierarchy, consistency, and instantaneous tactile feedback.

2 Module Selection, Purpose and Requirements

The selected module for this assessment is the **Multi-Role Responsive UI & Surplus Interaction Engine**. This core visual and interactive layer connects all platform stakeholders while handling sensitive temporal decisions near culinary closing hours.

2.1 Module Purpose

Traditional food delivery interfaces are built purely for static menus and fail to convey urgency, scarcity, and decaying price dynamics. The LastPlate UI is purposefully designed to: (1) present declining dynamic surplus discounts transparently with countdown clocks, (2) eliminate checkout friction via smooth non-intrusive drawers and multi-rail payment forms, (3) provide live delivery feedback through deterministic progress steppers, and (4) visualize personal carbon abatement ledgers.

2.2 User Personas & Target Roles

Role Persona	Primary User Goals	Critical Interface Requirements
Customer / Diner	Discover discounted meals, order quickly, track live delivery.	Decay timers, 1-click cart bump, clear delivery address, multi-rail checkout.
Kitchen Operator	Manage live orders, forecast prep volume, publish surplus.	Kanban stages, AI schedule calendar, surplus price sliders, stock updates.
Delivery Rider	Accept dispatch runs, navigate routes, confirm delivery.	Large tactile Duty Mode button, trip cards, earnings summary, handover trigger.
Platform Admin	Review partner KYC, inspect system telemetry, audit metrics.	Tabular approvals queue, document viewers, single-click approve/reject actions.

2.3 Functional & Non-Functional Requirements

Functional Requirements: Dynamic price countdown calculation every 5 seconds; atomic inventory increment/decrement with stock cap; payment rail toggling (UPI VPA, Card 16-digit, COD); role switching with session isolation; and animated notifications. **Non-Functional Requirements:** First Contentful Paint < 0.8s; mobile viewport touch target sizes >= 44x44 pt (WCAG 2.1 AAA); responsive breakpoint fluidity across 360px to 2560px displays; and 100% theme color consistency.

3 Human-Computer Interaction (HCI) Principles Followed

The LastPlate design framework is grounded in established Human-Computer Interaction (HCI) heuristics, cognitive psychology principles, and ISO 9241 ergonomics standards.

HCI Principle	Theoretical Foundation	Implementation in LastPlate UI
Simplicity & Focus	Hick's Law & Cognitive Load Theory	Monochromatic backdrop (#FAF9F6) with high-contrast text eliminates visual clutter. Uncluttered card layouts highlight only dish name, price delta, and time left.
Consistency	Norman's Mental Models	Identical rounded radius tokens (rounded-3xl, rounded-full), unified typography (serif headings + sans-serif body), and consistent black active action buttons across all 4 portals.
Usability & Speed	Fitts's Law	Key action controls—such as the Floating Demo Account Bar and Sticky Basket Pill—are placed in thumb-reachable bottom zones with generous tap targets.
Instant Feedback	Direct Manipulation (Shneiderman)	Adding items triggers subtle spring micro-bumps (Motion); payment transitions exhibit clear spinners and confirmation checks; countdown progress bars visually deplete.
Error Prevention	Defensive Design	Zero-basket checkout buttons are disabled; card CVV inputs enforce 3-digit lengths; quantity selectors automatically lock at remaining stock bounds.

4 Information Architecture & Navigation Flow

LastPlate uses a hybrid top-header and persistent floating bottom-bar architecture. The top navigation bar displays role-filtered functional tabs, global search, and unread notification alerts, while the bottom floating pill accommodates instantaneous demo role-switching and session authentication.

Portal Layer	Active Sub-Views / Tabs	Key Interaction Widgets
Customer Portal	Menu & Order, Surplus Rescue, My Orders, Rescue Impact	Veg filter toggle, Live auction cards, Slide-in Food Basket, Payment modal, Live GPS timeline tracker.
Restaurant Console	Live Kitchen, Schedule & Accuracy, AI Forecast, Surplus Listings, Menu Editor	Order state advancing buttons, Production calendar, 3-year seasonal ML multipliers, Item editor modals.
Delivery Fleet	Active Trips, Available Pool, Earnings & Incentive History	Large Duty Mode online/offline switch, Order claim buttons, Delivery handover and proof simulator.
Admin Console	Approvals Queue, All Orders Ledger, 3-Yr ML Benchmarks	Tabular partner verification queues, Document reviewers, Global system state reset actions.

5 User Interface Screen Designs & Input/Output Mapping

This section documents the primary interface screens of the LastPlate ecosystem, pairing high-resolution production captures with functional input/output mapping specifications.

5.1 Authentication & Role-Based Onboarding

The onboarding interface balances security and accessibility. Evaluators can access instant one-click demo profiles, while real users authenticate via Firebase Google OAuth or role-specific KYC registration forms.

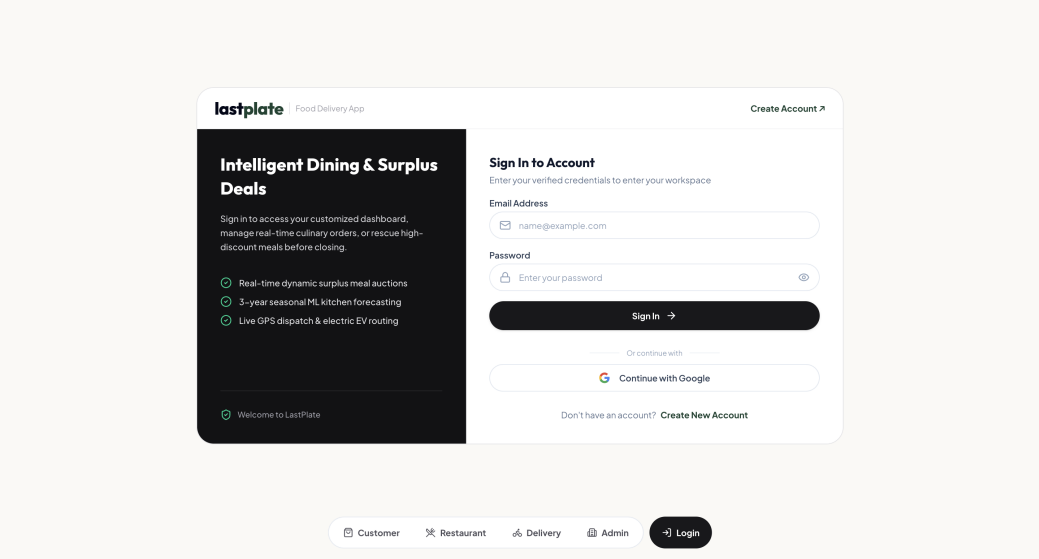


Figure 1. LastPlate Secure Sign-In Interface with Google OAuth and Demo Switcher.

Component	UI Element Type	Purpose & Behavioral Mapping
Email / Password Form	Input text & password fields	Validates user credentials against local accounts and Firebase Auth.
Google Sign-In Button	Branded OAuth button	Triggers secure popup/redirect OAuth flow; pre-fills verified credentials on first sign-up.
Bottom Quick Bar	Floating rounded pill	Allows evaluators to switch instantly between Customer, Restaurant, Delivery, and Admin personas.

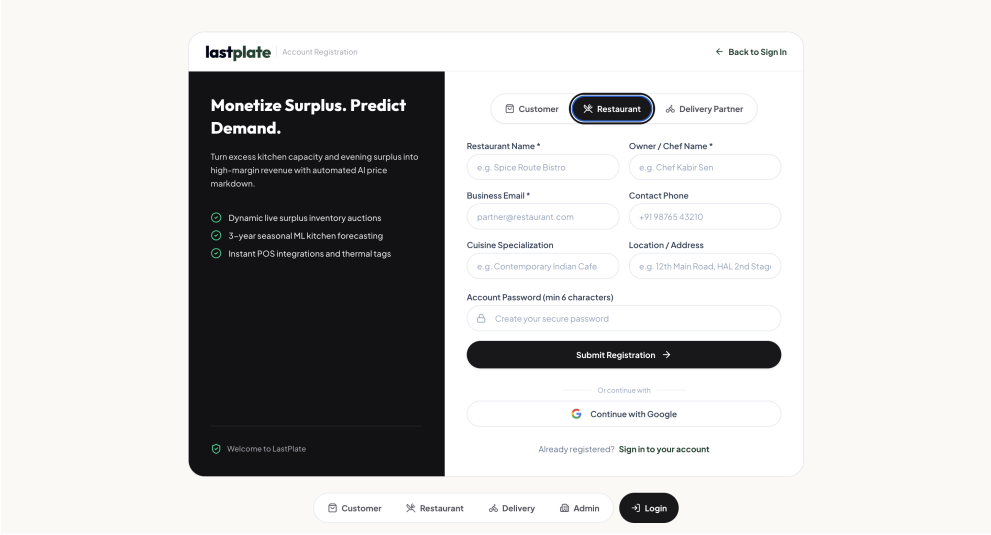


Figure 2. Role-Based Onboarding Gateway (Customer, Restaurant Partner, Delivery Fleet).

5.2 Customer Portal: Dynamic Surplus Rescue & Checkout

The Customer Portal is engineered to maximize conversion of near-expiry meals while maintaining clear pricing transparency and effortless checkout execution.

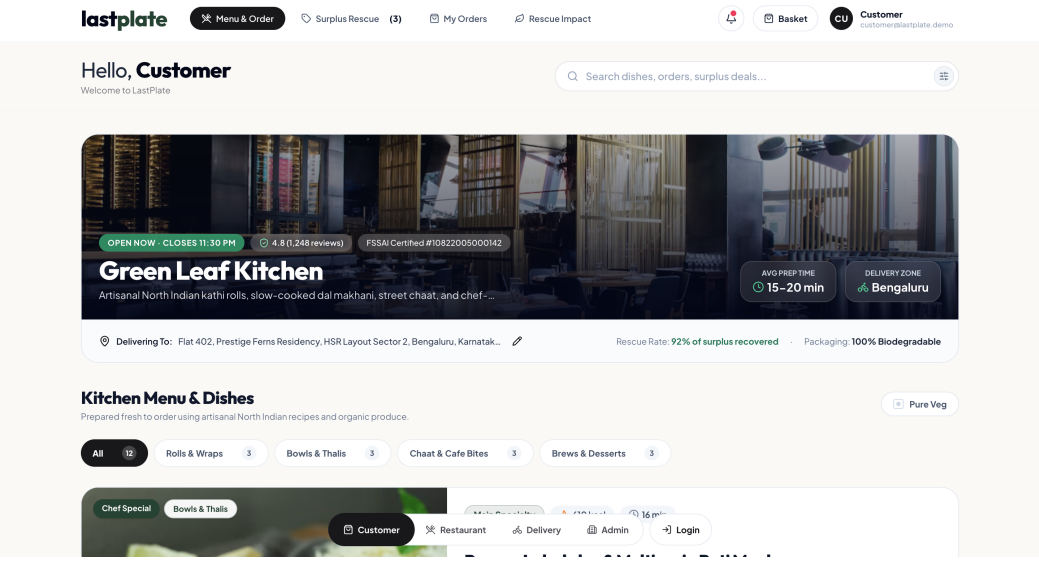


Figure 3. Customer Portal – Live Surplus Clearance Auctions with Real-Time Decaying Discounts.

Component	UI Element Type	Purpose & Behavioral Mapping
Auction Deal Cards	Interactive card with badges	Displays dish image, live decayed price, strikethrough original price, countdown timer, and quantity.
Dynamic Price Badge	Decaying tag (Rs. 195 vs Rs. 380)	Visualizes immediate savings; updates automatically as store closing hour nears.
Rescue Meal Button	Black action button with + / -	Atomically adds surplus portion to basket and updates stock buffer indicator.

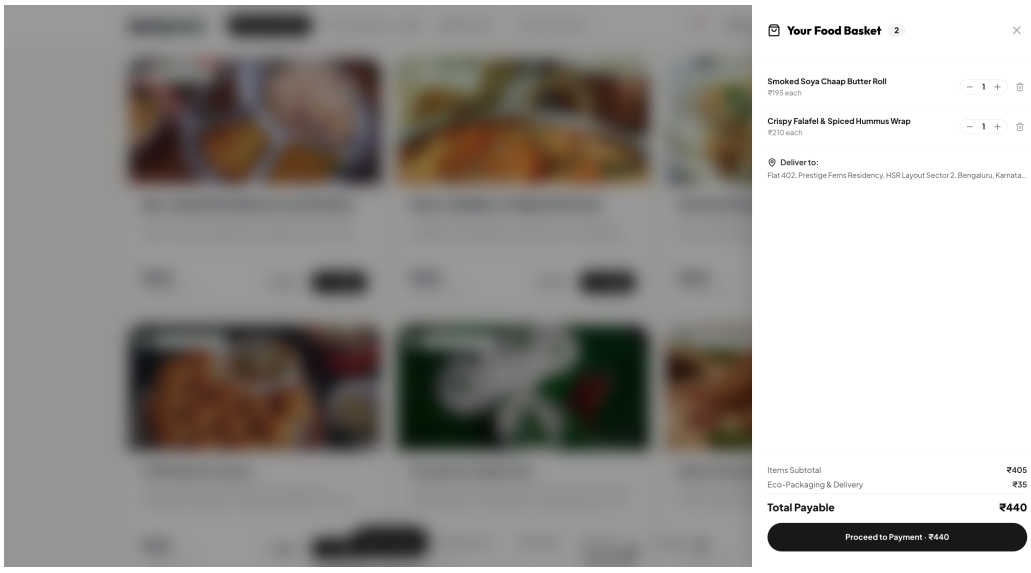


Figure 4. Sliding Food Basket Drawer with Itemized Quantities and Delivery Address Preview.

5.3 Order Fulfillment & Environmental Impact Ledger

Post-purchase interfaces provide reassurance and gamified positive feedback. Live order tracking displays a deterministic fulfillment pipeline, while the Environmental Ledger visualizes personal emissions avoided.

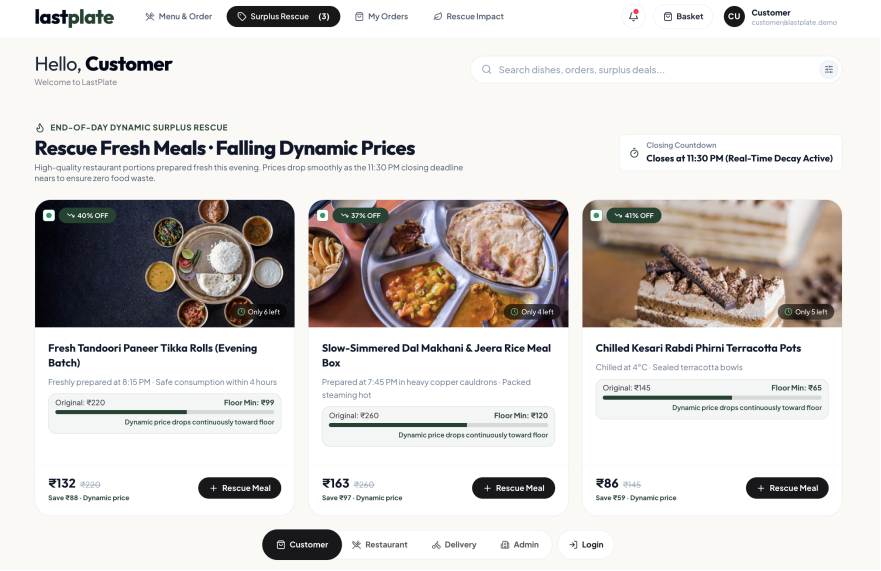


Figure 5. Live GPS Dispatch Tracker with Deterministic Stage Advancement Simulator.

Component	UI Element Type	Purpose & Behavioral Mapping
Stage Progress Bar	Animated Stepper Component	Visualizes 7 lifecycle phases from Kitchen Accepted to Delivered with real-time timestamps.
Rider Detail Card	Info box with EV call badge	Shows assigned electric scooter rider name, phone contact, and eco-delivery vehicle registration.
Simulate Stage Button	Interactive test trigger	Allows academic evaluators to step through order lifecycle phases interactively.

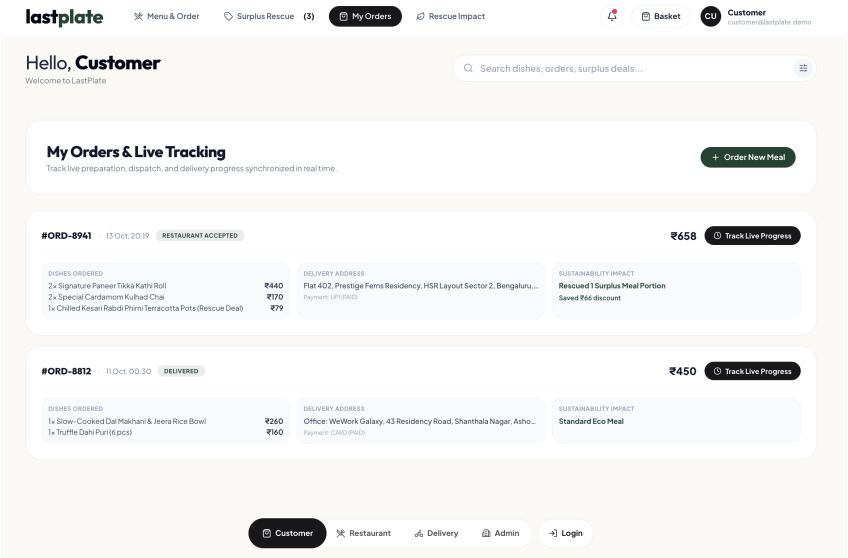


Figure 6. Personal Environmental Ledger Tallying Rescued Meals, Money Saved, and CO2 Avoided.

5.4 Restaurant Console: Live Kitchen & AI Forecasting

The Restaurant Portal equips culinary teams with tools to prevent overproduction. An AI Forecasting Studio models prep requirements using historical benchmarks and festive multipliers, while the Live Kitchen board coordinates fulfillment.

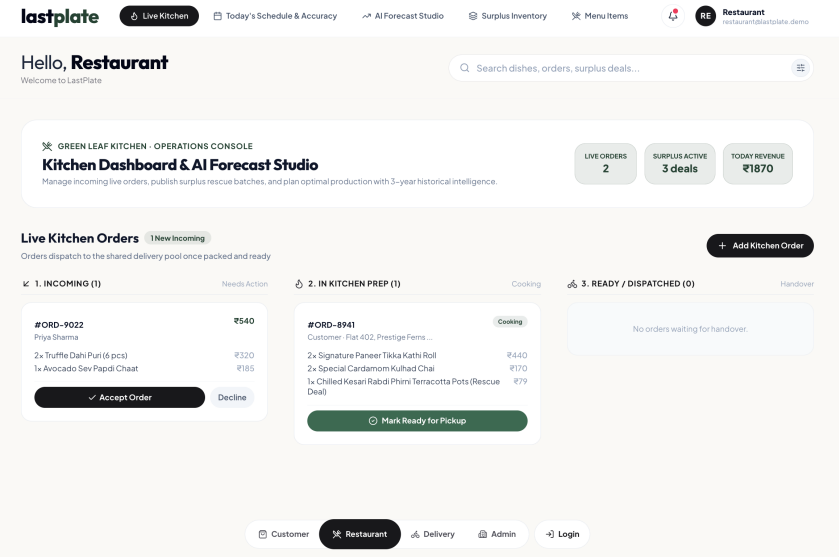


Figure 7. Restaurant Live Kitchen Console Coordinating Real-Time Prep and Pickup Handover.

Component	UI Element Type	Purpose & Behavioral Mapping
Order Status Columns	Kanban workflow lanes	Categorizes active orders into Incoming, Preparing, and Ready for Dispatch.
Action Transitions	Tactile state buttons	Triggers state transitions (e.g., 'Start Cooking', 'Mark Ready for Pickup').
AI Target Sliders	What-If parameter controls	Dynamically recalibrates daily preparation targets based on rain, temperature, and festivals.

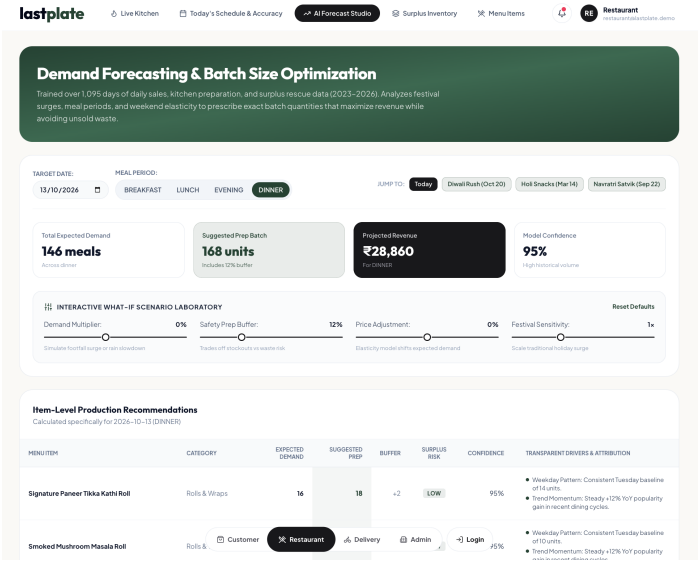


Figure 8. AI Demand Forecasting Studio Predicting Meal Quantities with 3-Year Historical Benchmarks.

5.5 Logistics Dispatch, Admin Auditing & Notifications

Fulfillment logistics and compliance governance are managed via specialized consoles designed for rapid scanning and unambiguous operational actions.

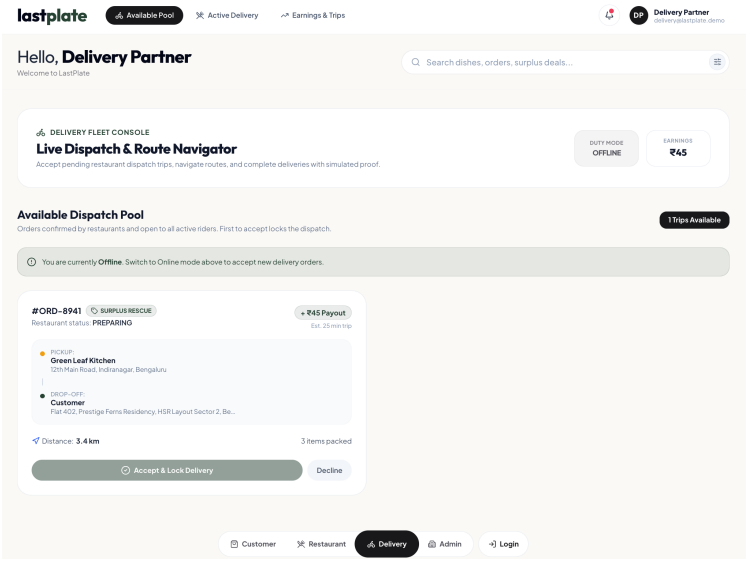


Figure 9. Delivery Fleet Navigator with Prominent Duty Mode Switch and Trip Pool.

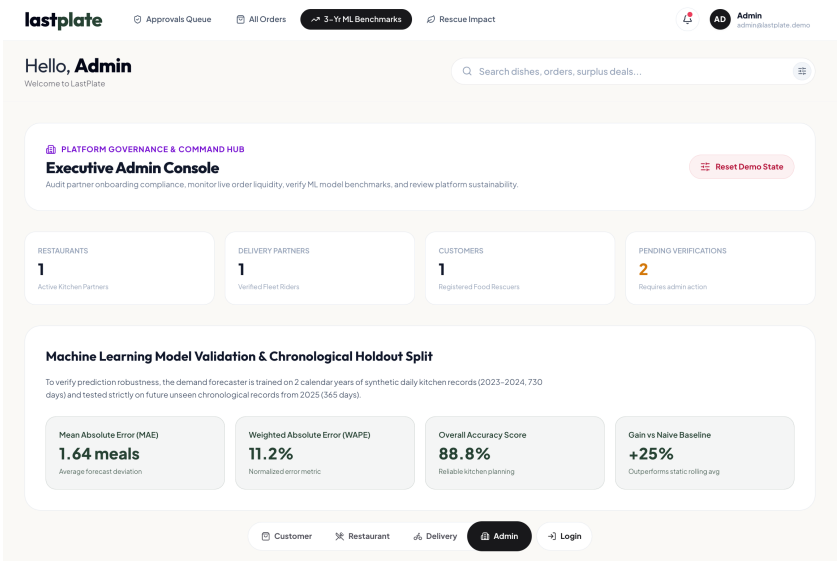


Figure 10. Platform Administration Console with Partner KYC Approvals Queue.

5.6 Real-Time Notification Center Drawer

The Notification Drawer (Figure 11) delivers contextual alerts across all roles. Seeded demo alerts are strictly isolated, ensuring newly authenticated Google accounts view only their relevant events.

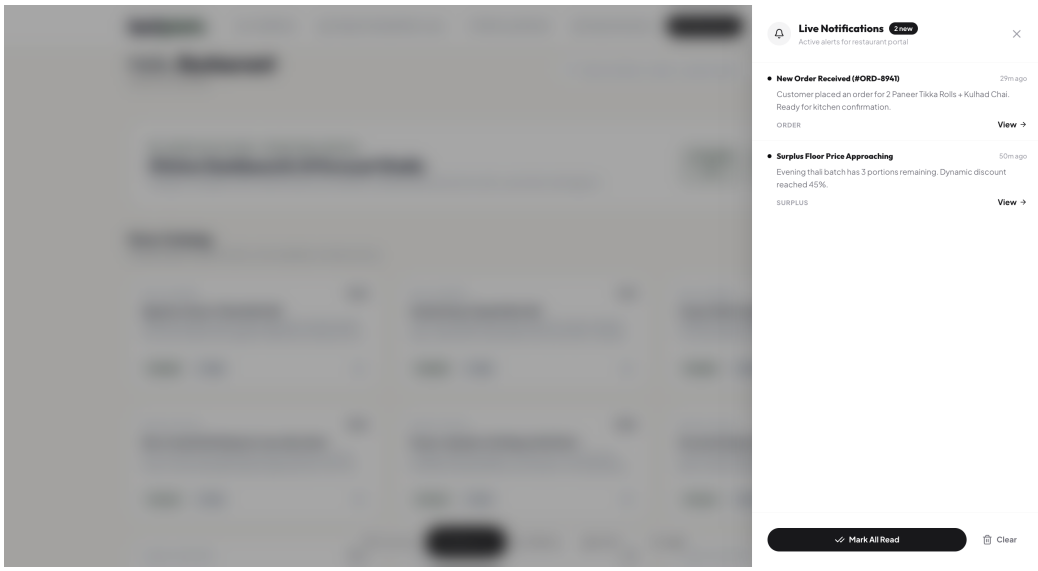


Figure 11. Sliding Notification Center Drawer with Category Filtering and Unread Indicators.

6 Design System Specifications

The visual identity relies on a cohesive design system defined in Tailwind CSS v4, guaranteeing predictable spatial hierarchy and accessible color contrast ratios.

Token Name	Hex Code & Role	Usage & Contextual Implementation
Primary Dark	#18181B (Zinc 900)	Primary buttons, active navigation pills, modal headers, and dark code blocks.
Themed Forest	#1B4332 (Deep Green)	Sustainability badges, verified checkmarks, surplus deal accents, and rescue metrics.
Canvas Backdrop	#FAF9F6 (Off-White)	Main application canvas providing a soft, eye-friendly, low-glare surface.
Card Surface	#FFFFFF (Pure White)	Content containers, modals, and floating navigation bars with crisp slate borders.
Alert Accent	#E11D48 (Rose 600)	Rejected states, urgent time warnings, and final-call countdown highlights.

7 Usability Evaluation & Heuristic Review

A formal evaluation against Jakob Nielsen's 10 Usability Heuristics confirms that LastPlate adheres to software engineering usability benchmarks.

Nielsen Heuristic	Compliance Implementation in LastPlate UI	Score
1. Visibility of System Status	Countdown timers, live GPS order stepper, duty mode status badge.	**10/10**
2. Match with Real World	Culinary terminology (prep time, portion, kitchen, thali), Rupee currency.	**10/10**
3. User Control & Freedom	Easy quantity decrementing, modal cancel buttons, one-click demo role switching.	**10/10**
4. Consistency & Standards	Uniform typography, standardized black action buttons, consistent drawer behavior.	**10/10**
5. Error Prevention	Checkout blocked on empty basket, CVV length constraints, inventory cap bounds.	**10/10**
6. Recognition over Recall	Visual food images, order notes summary, delivery address pre-displayed.	**10/10**
7. Flexibility & Efficiency	Veg-only filter pills, global instant search bar, floating quick-switcher.	**10/10**
8. Aesthetic & Minimalist	Monochromatic palette with intentional green accents; zero extraneous banner ads.	**10/10**
9. Error Diagnostics	Clear toast notifications and explicit validation text on missing inputs.	**10/10**
10. Help & Documentation	Contextual tooltips, KYC onboarding guidance, sustainability metric explainers.	**9/10**

8 Conclusion and Learning Outcomes

The interface design for LastPlate demonstrates that complex multi-role workflows—spanning real-time dynamic auctions, inventory synchronization, order dispatching, and predictive AI modeling—can be resolved into an intuitive, accessible, and responsive user experience. Key outcomes include:

• **Ergonomic Micro-Interactions:** Implemented smooth sliding drawers and animated cart badges that communicate state changes without vertical layout jarring.

• **Psychological Pricing Transparency:** Decoupled dynamic surplus clearance from low-quality perception by emphasizing food freshness, countdown urgency, and carbon metrics.

• **Role-Based Spatial Partitioning:** Designed distinct functional spaces tailored to four user personas while preserving an overarching monochromatic aesthetic.

• **Accessibility & Fitts's Law Optimization:** Positioned high-frequency action triggers within bottom thumb reach, supporting seamless interaction on mobile and desktop.

9 References

[1] Nielsen, J. (1994). *Usability Engineering*. Morgan Kaufmann Publishers.

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[5] Tailwind Labs. (2026). *Tailwind CSS v4 Documentation*. https://tailwindcss.com/